

From LLMs to reasoning models

Agentic AI, part 1

Carlos Cotrini

September 2026

1

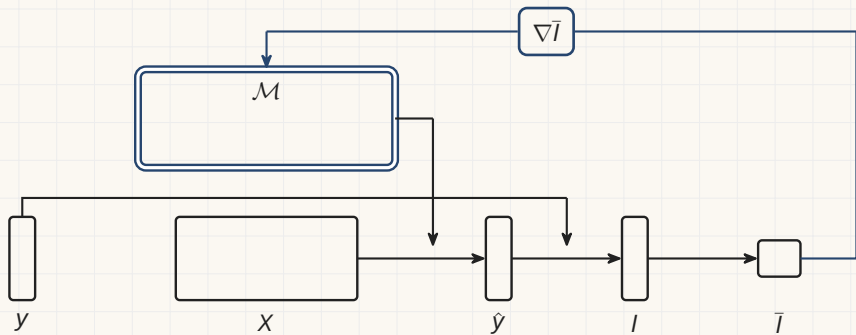
Section 1

How a language model is trained

The loop, with the model as one box

The picture from weekend 1, with the network as one box: the model $\mathcal{M}$.

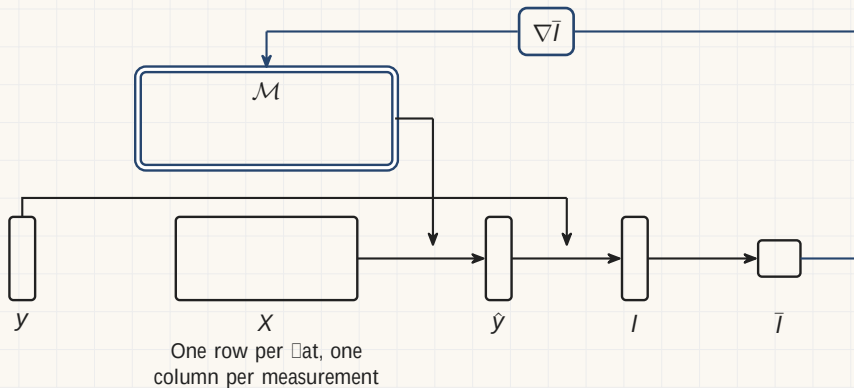
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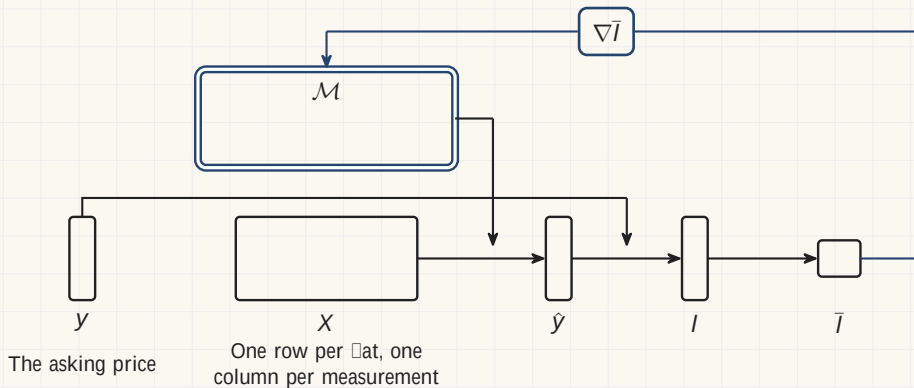
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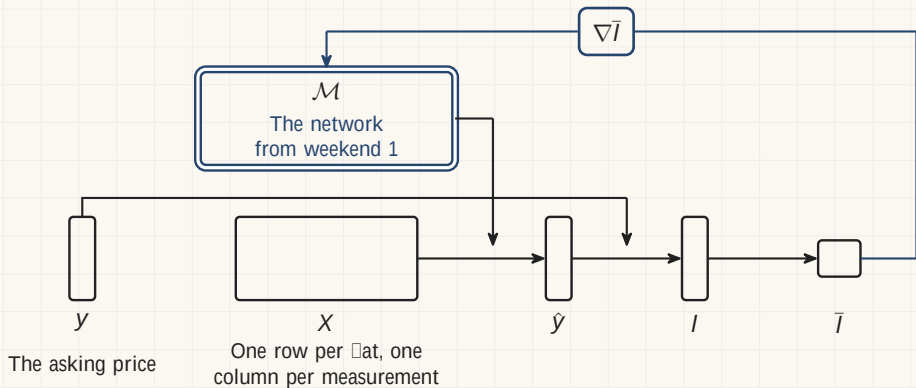
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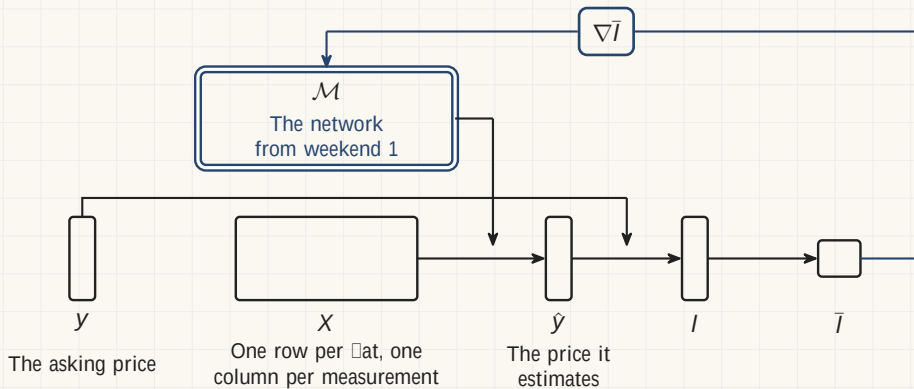
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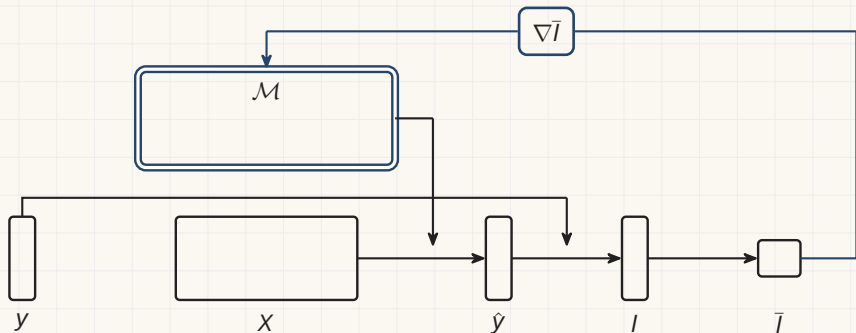
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Large language models

A row is text with its next token cut off. Every position of every document gives one.

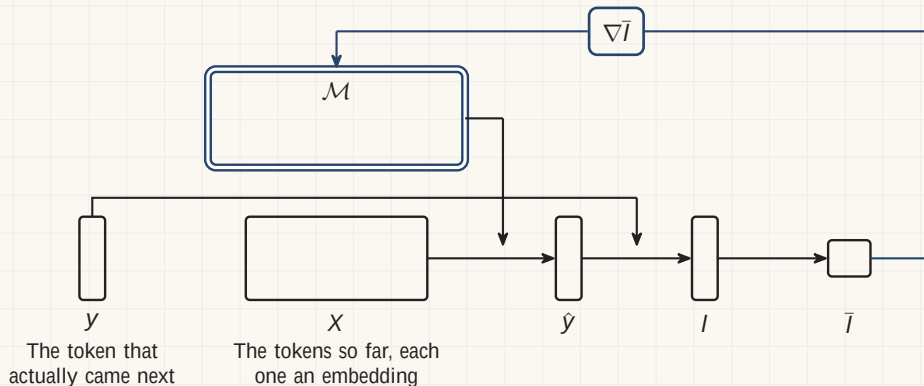
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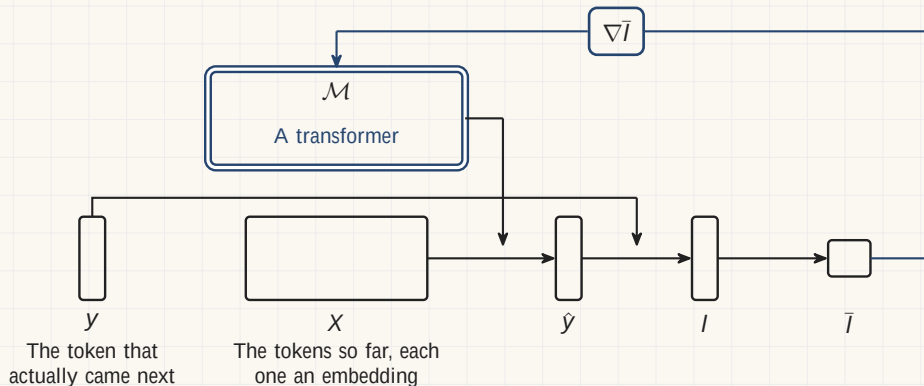
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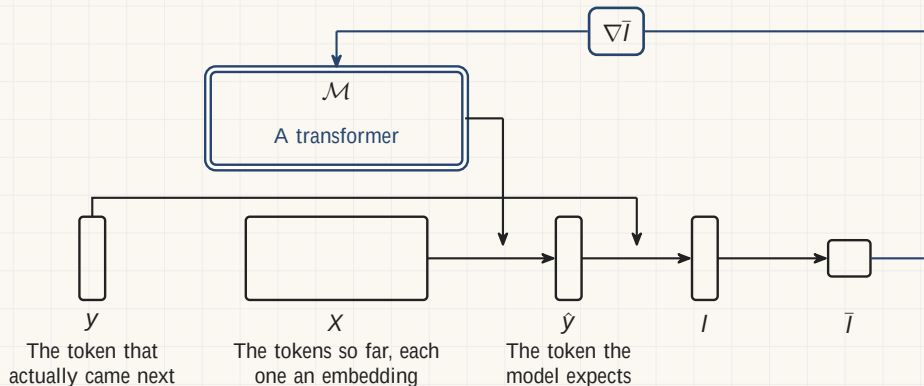
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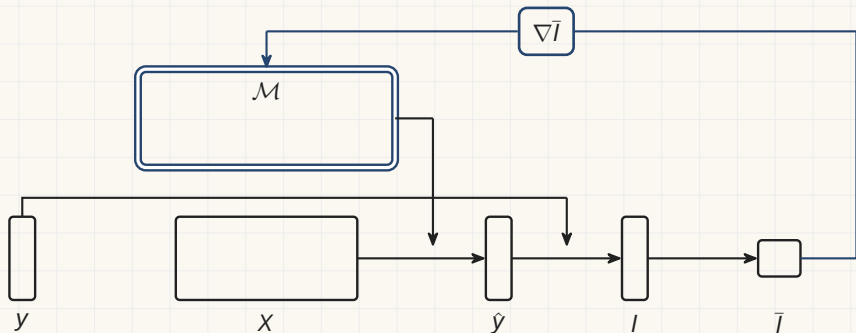
Section 2

How a language model becomes a chatbot

Instruction tuning

GPT-3 was $\square$ nished in 2020 and it completed text. A smaller second pass made it answer.

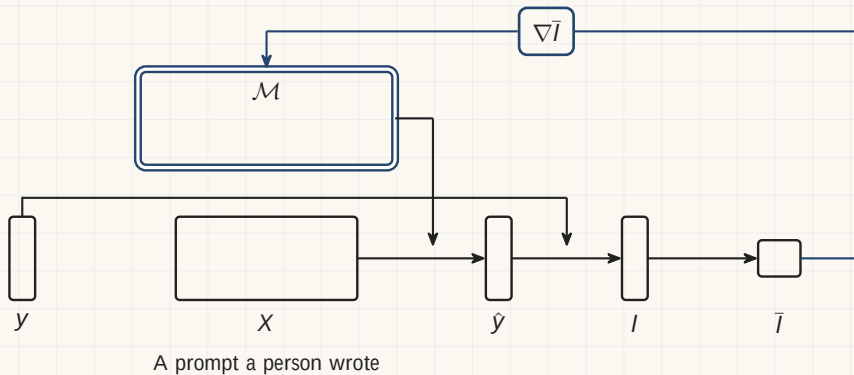
12 725 training prompts, and its replies were preferred to a model 100 times larger. Ouyang and others, 2022.



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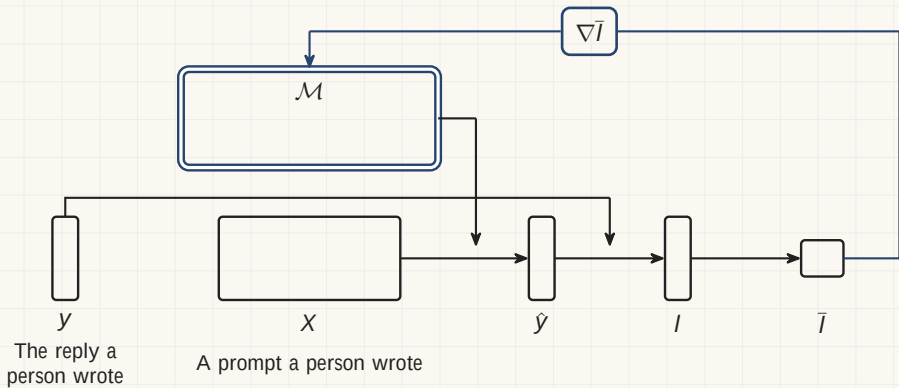
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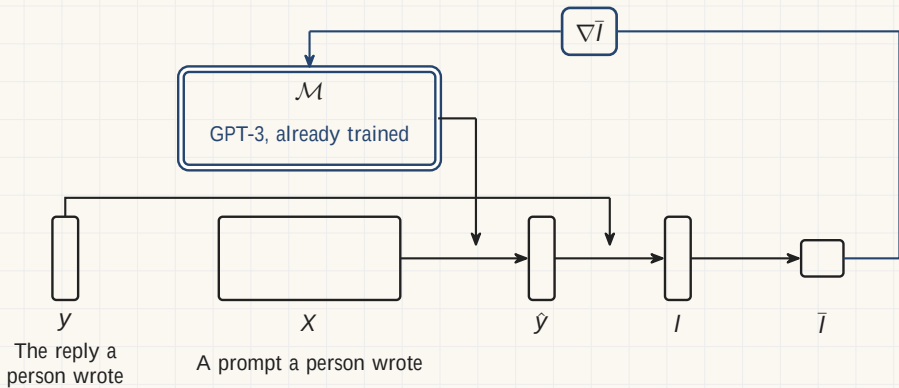
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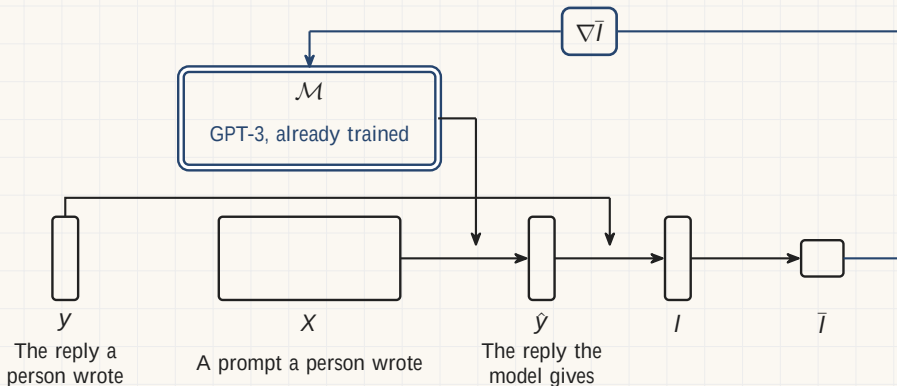
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The instruction tuning data

About 40 contractors, hired on Upwork and through Scale AI, wrote the replies and did the ranking.

Example from Figure 2, counts from table 6. Ouyang and others, 2022.

One demonstration

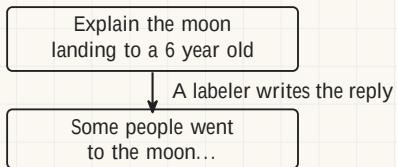
Explain the moon
landing to a 6 year old

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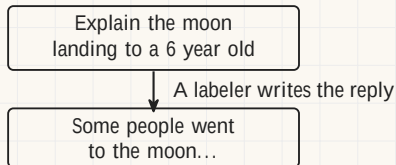
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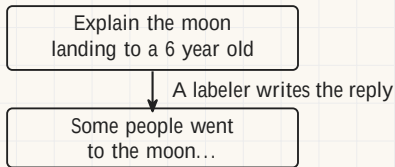
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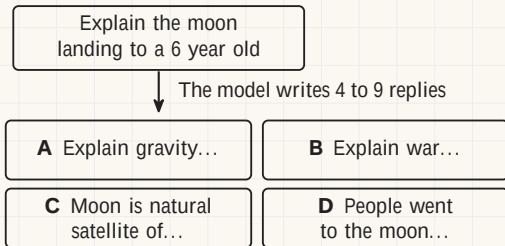
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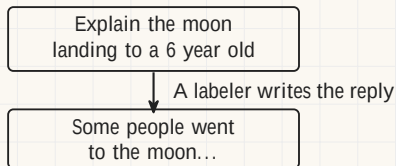
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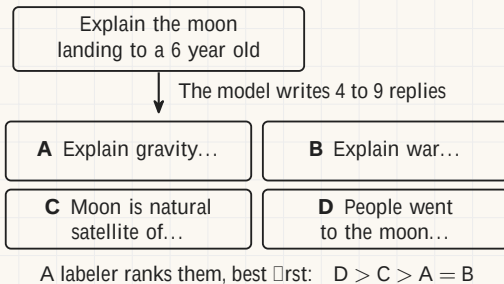
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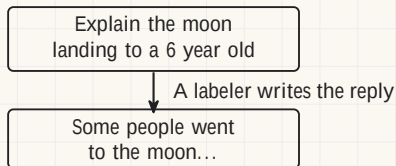
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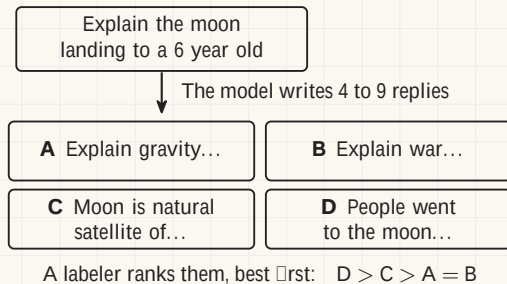
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33 207 prompts, most of them sent by users of the API



Game of 24



$$\begin{array}{r} 4 + 6 + 2 - 8 \\ \hline \end{array}$$

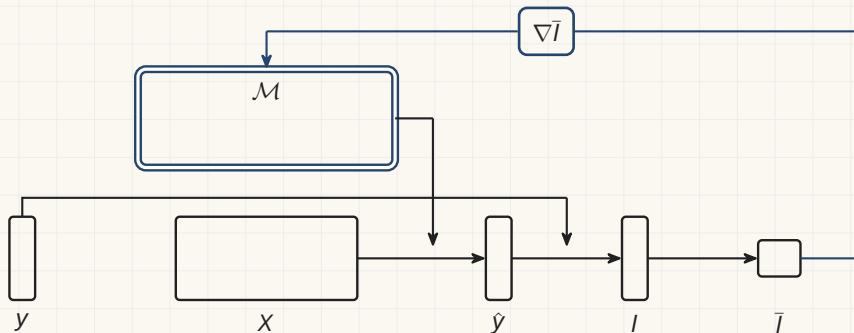
Section 3

How a chatbot becomes a reasoning model ²⁴

Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

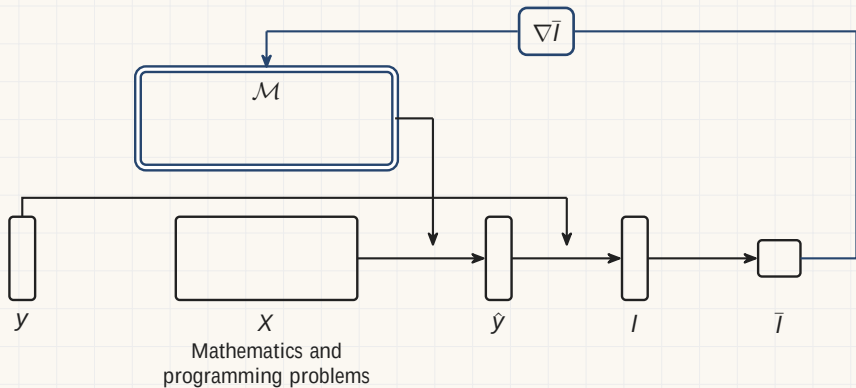
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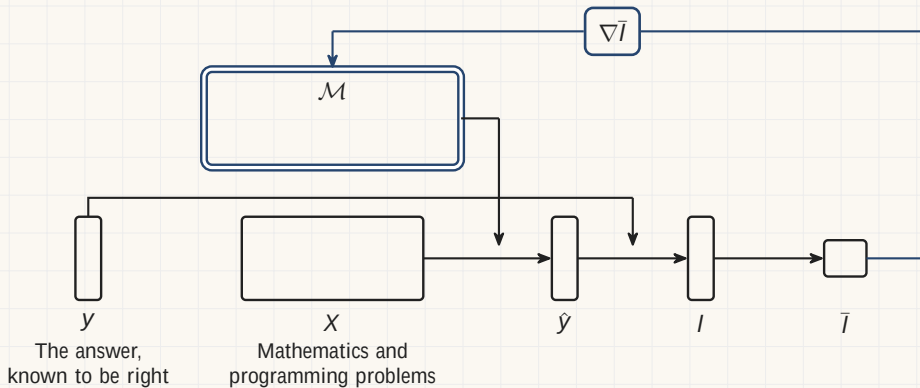
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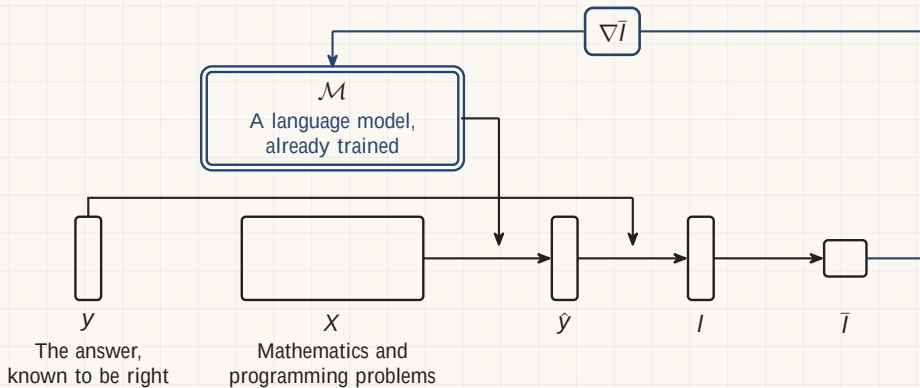
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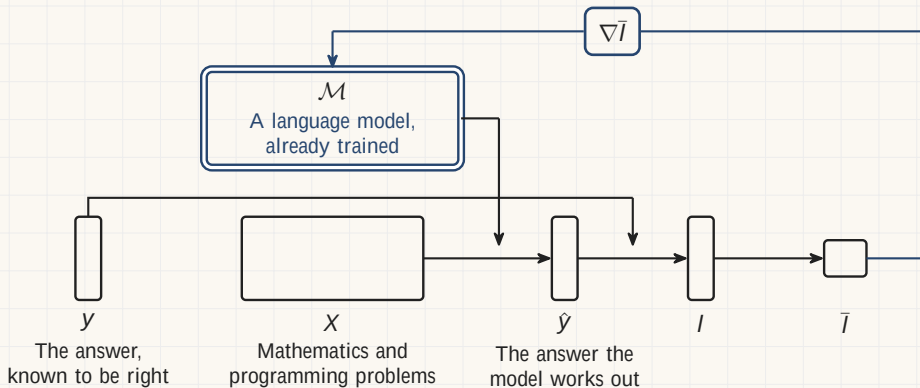
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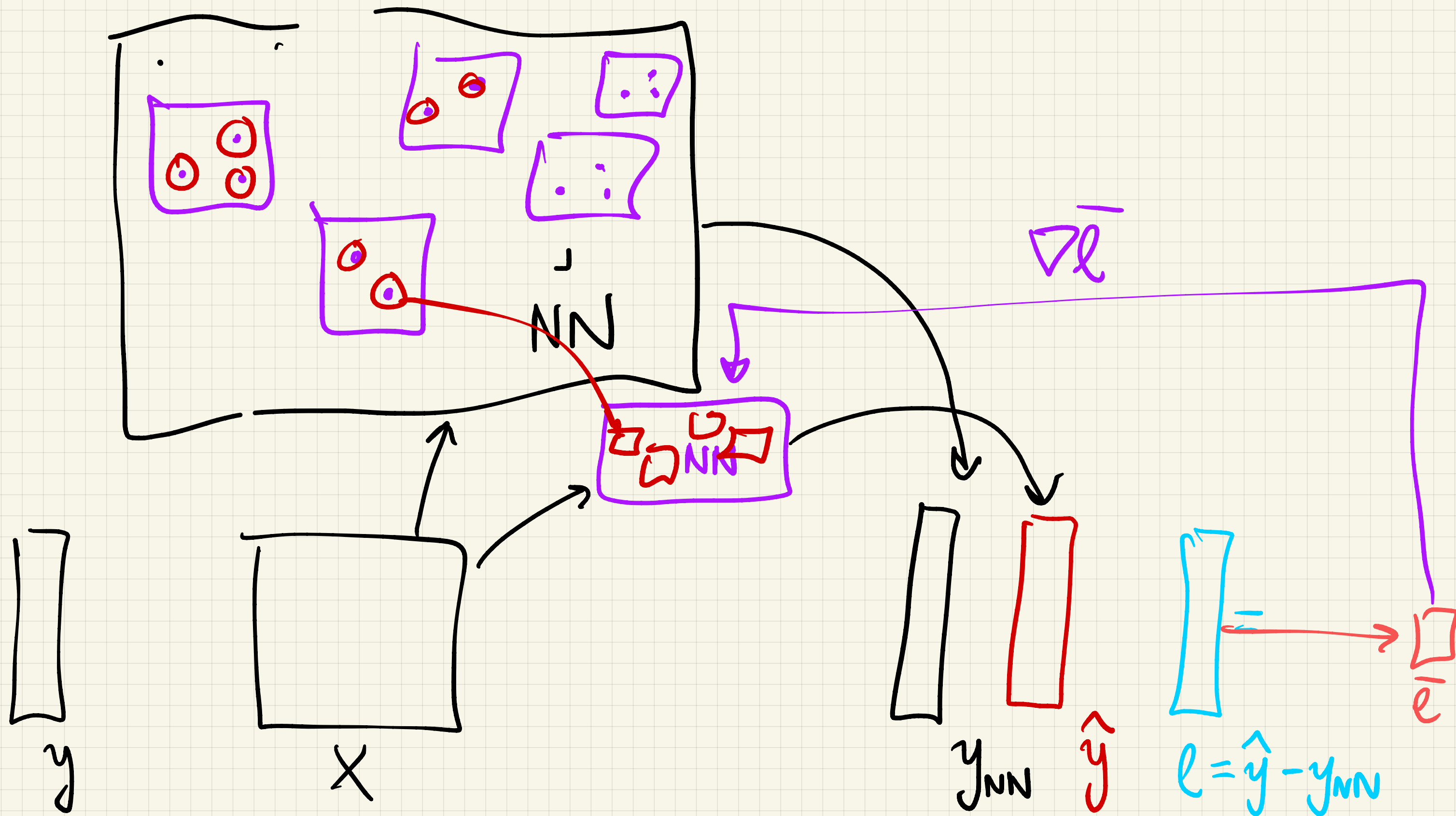
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Distillation



4

Section 4

Reinforcement learning

A language model as a Markov decision process

Writing an answer is a chain of decisions, one token at a time.

Words stand in for tokens. Liu and others, *Understanding R1-Zero-like training*, 2025.

What is 12 times 7?

12

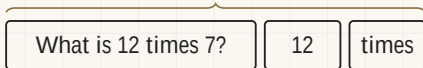
times

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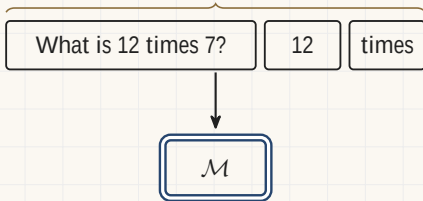


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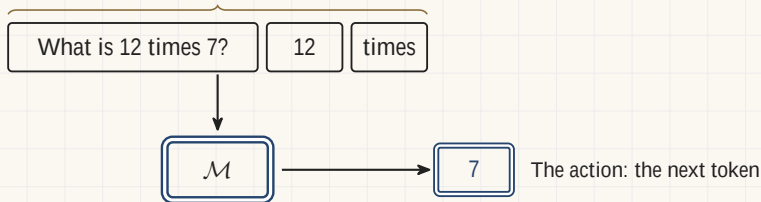


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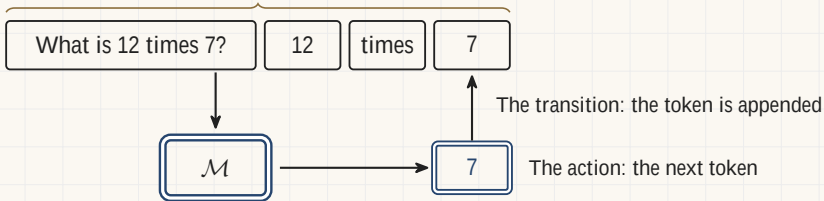


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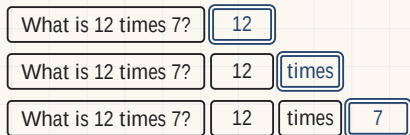
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The end of the episode, and the reward

The model writes until it chooses the end-of-answer token. Then a program checks the answer.

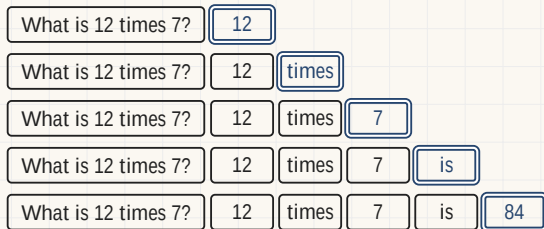
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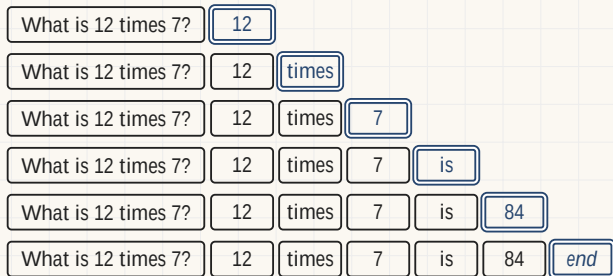
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The end-of-answer token: the episode ends

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						Reward
What is 12 times 7?	12					0
What is 12 times 7?	12	times				0
What is 12 times 7?	12	times	7			0
What is 12 times 7?	12	times	7	is		0
What is 12 times 7?	12	times	7	is	84	0
What is 12 times 7?	12	times	7	is	84	end

0 while the answer is still being written

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What is 12 times 7?	12	times	7	is	84	1
					end	
The end-of-answer token: the episode ends						

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1 at the end: 84 is right.
A wrong answer earns 0

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						1 The sum of the rewards

Gradient descent, from weekend 1

Weekend 1 made a loss smaller by stepping against its tilt.

L is the loss, for example the average squared miss over the houses.

W ← W
⏟
The weights

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$$\underbrace{W}_{\text{The weights}} \leftarrow W - \underbrace{\alpha}_{\text{Minus a small step}}$$

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Gradient ascent on the sum of rewards

Here the step goes with the tilt, to make the expected sum of rewards larger.

Policy gradient methods. Sutton and Barto, *Reinforcement Learning*, 2nd edition, 2018, chapter 13.

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References

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